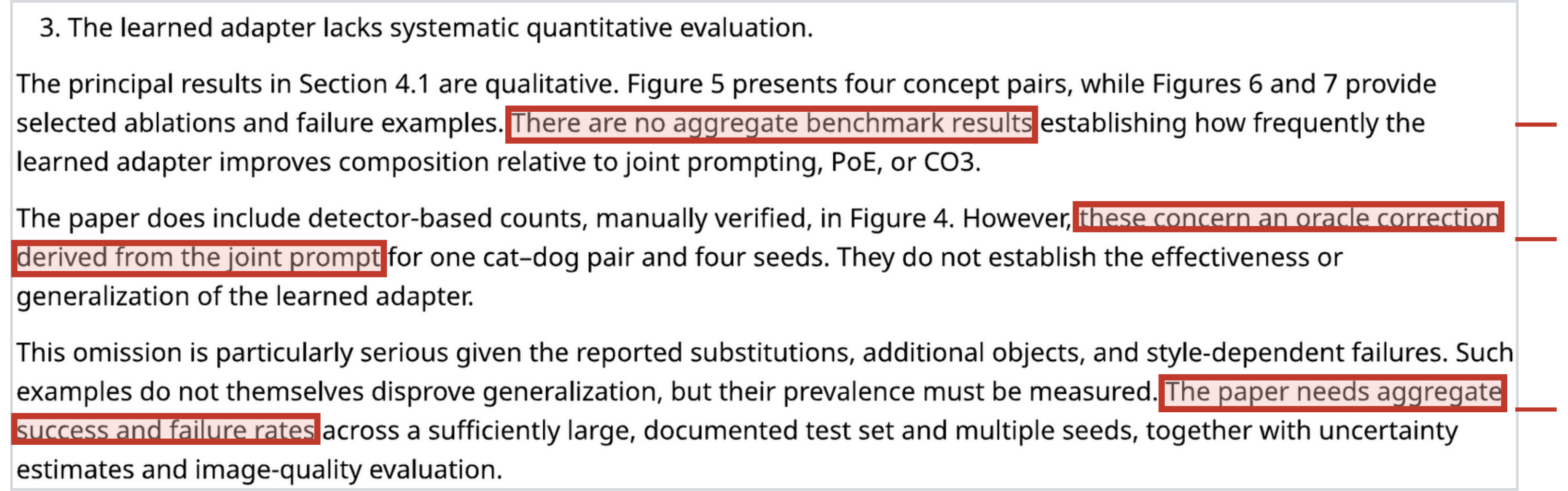
G3: How often the adapter works on unseen pairs

k4nd weakness 3 and question 2, UHz6 weakness 1 and question 2 · status: answered

THE REVIEWER'S POINT

k4nd, weakness 3



3. The learned adapter lacks systematic quantitative evaluation.

The principal results in Section 4.1 are qualitative. Figure 5 presents four concept pairs, while Figures 6 and 7 provide selected ablations and failure examples. There are no aggregate benchmark results establishing how frequently the learned adapter improves composition relative to joint prompting, PoE, or CO3.

The paper does include detector-based counts, manually verified, in Figure 4. However, these concern an oracle correction derived from the joint prompt for one cat–dog pair and four seeds. They do not establish the effectiveness or generalization of the learned adapter.

This omission is particularly serious given the reported substitutions, additional objects, and style-dependent failures. Such examples do not themselves disprove generalization, but their prevalence must be measured. The paper needs aggregate success and failure rates across a sufficiently large, documented test set and multiple seeds, together with uncertainty estimates and image-quality evaluation.

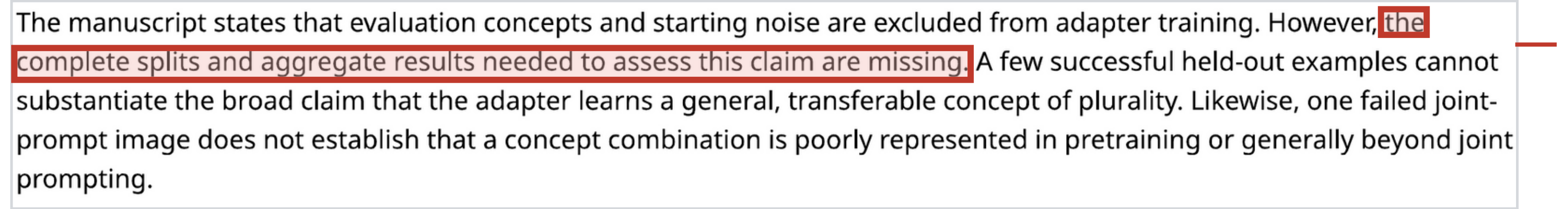
K-W3a

K-W3b

K-W3c

Reviews.pdf, page 4 (no text layer; passage located by OCR)

k4nd, weakness 3 (continued)

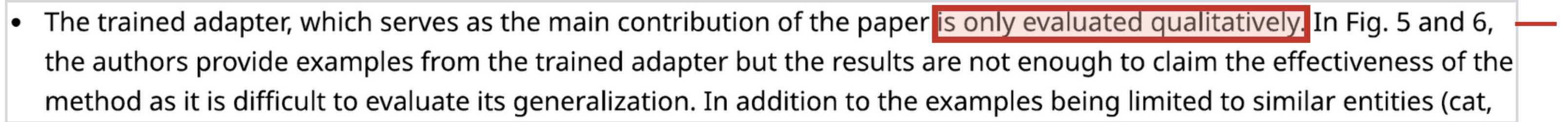


The manuscript states that evaluation concepts and starting noise are excluded from adapter training. However, the complete splits and aggregate results needed to assess this claim are missing. A few successful held-out examples cannot substantiate the broad claim that the adapter learns a general, transferable concept of plurality. Likewise, one failed joint-prompt image does not establish that a concept combination is poorly represented in pretraining or generally beyond joint prompting.

K-W3d

Reviews.pdf, page 5 (no text layer; passage located by OCR)

UHz6, weakness 1 and question 2

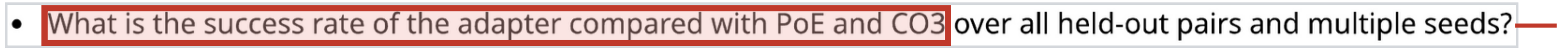


- The trained adapter, which serves as the main contribution of the paper is only evaluated qualitatively. In Fig. 5 and 6, the authors provide examples from the trained adapter but the results are not enough to claim the effectiveness of the method as it is difficult to evaluate its generalization. In addition to the examples being limited to similar entities (cat,

U-W1a

Reviews.pdf, page 3 (no text layer; passage located by OCR)

UHz6, question 2



- What is the success rate of the adapter compared with PoE and CO3 over all held-out pairs and multiple seeds?

U-Q2

Reviews.pdf, page 3 (no text layer; passage located by OCR)

these counts

this claim

THE SUBMITTED WEAKNESS

Figure 4, the only counts

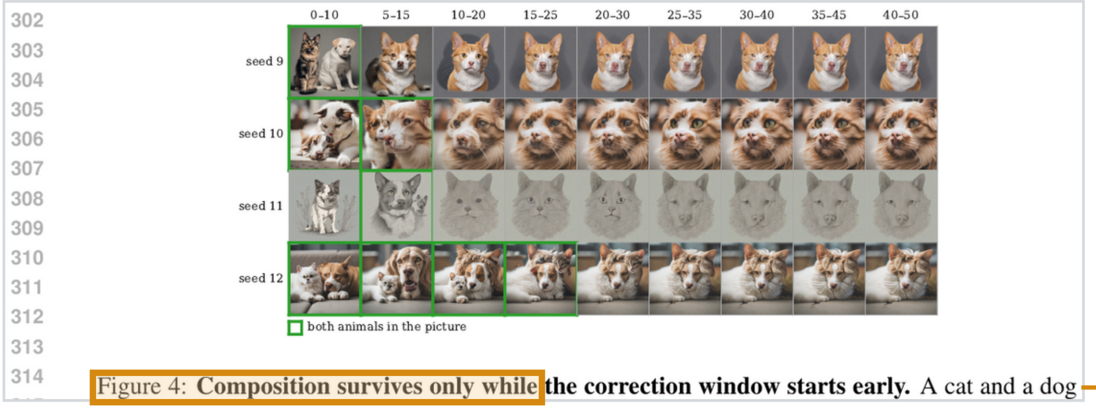
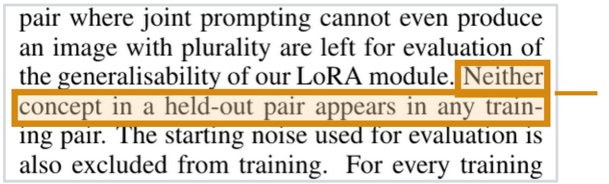


Figure 4: **Composition survives only while the correction window starts early.** A cat and a dog

oracle correction, one pair

56173_Restoring_plurality_in_c.pdf (submitted), page 6

Section 4, the held-out claim



pair where joint prompting cannot even produce an image with plurality are left for evaluation of the generalisability of our LoRA module. Neither concept in a held-out pair appears in any training pair. The starting noise used for evaluation is also excluded from training. For every training

no concept shared?

56173_Restoring_plurality_in_c.pdf (submitted), page 7

learned adapter, 53 pairs

failure rates

protocol stated

THE REVISED ANSWER

Table 3, 53 held-out pairs

held-out success, 95% CI

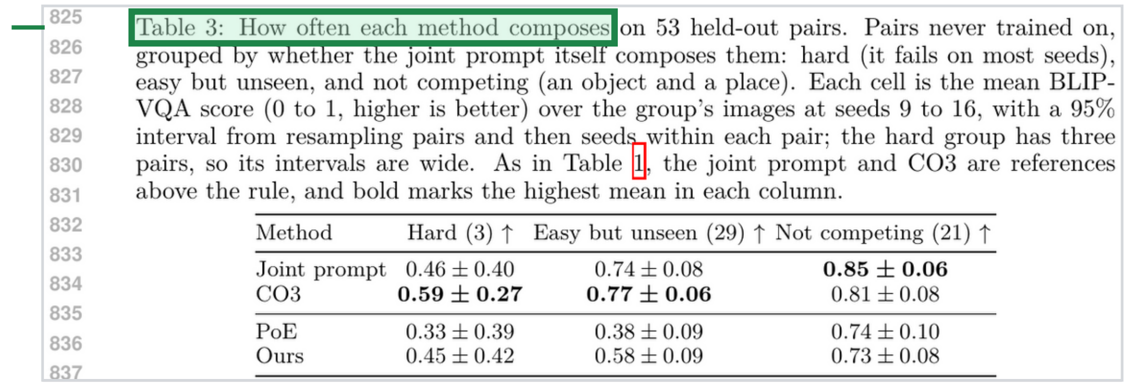
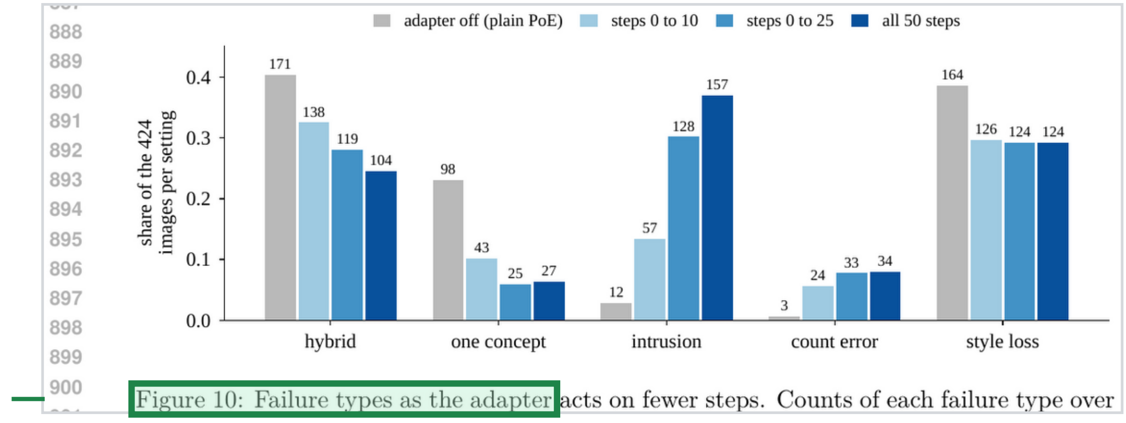


Table 3: How often each method composes on 53 held-out pairs. Pairs never trained on, grouped by whether the joint prompt itself composes them: hard (it fails on most seeds), easy but unseen, and not competing (an object and a place). Each cell is the mean BLIP-VQA score (0 to 1, higher is better) over the group's images at seeds 9 to 16, with a 95% interval from resampling pairs and then seeds within each pair; the hard group has three pairs, so its intervals are wide. As in Table 1, the joint prompt and CO3 are references above the rule, and bold marks the highest mean in each column.

Method	Hard (3) ↑	Easy but unseen (29) ↑	Not competing (21) ↑
Joint prompt	0.46 ± 0.40	0.74 ± 0.08	$\mathbf{0.85 \pm 0.06}$
CO3	$\mathbf{0.59 \pm 0.27}$	$\mathbf{0.77 \pm 0.06}$	0.81 ± 0.08
PoE	0.33 ± 0.39	0.38 ± 0.09	0.74 ± 0.10
Ours	0.45 ± 0.42	0.58 ± 0.09	0.73 ± 0.08

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 16

Figure 10, failure types counted



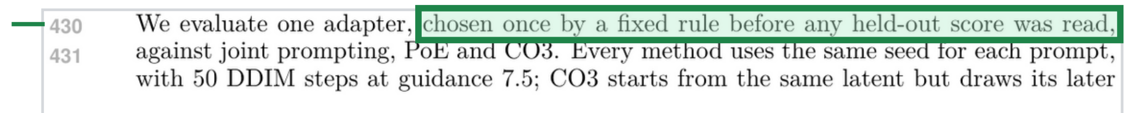
failures counted

Figure 10: Failure types as the adapter acts on fewer steps. Counts of each failure type over

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 17

Section 4.1, the protocol

fixed before scoring



We evaluate one adapter, chosen once by a fixed rule before any held-out score was read, against joint prompting, PoE and CO3. Every method uses the same seed for each prompt, with 50 DDIM steps at guidance 7.5; CO3 starts from the same latent but draws its later

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 8

The reply we plan to send (replies/k4nd-and-uhz6.md: k4nd W3 and Q2, UHz6 Q2)

answered

W3: no systematic quantitative evaluation of the adapter

We agree, and the revision adds this evaluation. The adapter, the seeds and the success bar were all fixed before any held-out score was read (Section 4.1).

Table 1 scores every method on the 276 Attend-and-Excite prompts (66 animal pairs, 144 animal-object pairs, 66 object pairs) at seeds 9 to 12, with composition measured by BLIP-VQA and image quality by CLIP score and ImageReward.

Table 3 scores the 53 held-out pairs described in Appendix C.2, grouped as hard, easy but unseen, and not competing, at seeds 9 to 16, with 95% intervals from resampling pairs.

The training and held-out pairs, with every seed, are released with the code [TODO: anonymous link]; Appendix C.2 describes the held-out groups. The submission said that no held-out concept appears in training; that was too strong, and Section 4 now says the held-out pairs are unseen as pairs.

The failure types the reviewer names (substituted identities, additional objects, style-dependent loss) are counted per setting in Figure 10 (Appendix C.3).

Q2: aggregate results over held-out concepts and seeds

Answered under k4nd W3: Tables 1 and 3. The identity errors, additional objects and style-dependent failures are counted per setting in Figure 10 (Appendix C.3).

Q2: success rate against PoE and CO3 over all held-out pairs and seeds

Answered under k4nd W3: Tables 1 and 3.

Changes behind this page: Table 3 (53 held-out pairs, seeds 9 to 16, 95% intervals), Figure 10 (failure types counted), Section 4.1 opens with the protocol; Section 4 says held-out pairs are unseen as pairs.

UHz6 weakness 1 also asks for standard benchmarks; that half is on the G4 page. The reply's anonymous code link is still a TODO. Every excerpt is a crop of a page render; the file and page are printed under it. Boxes mark text already on the page.